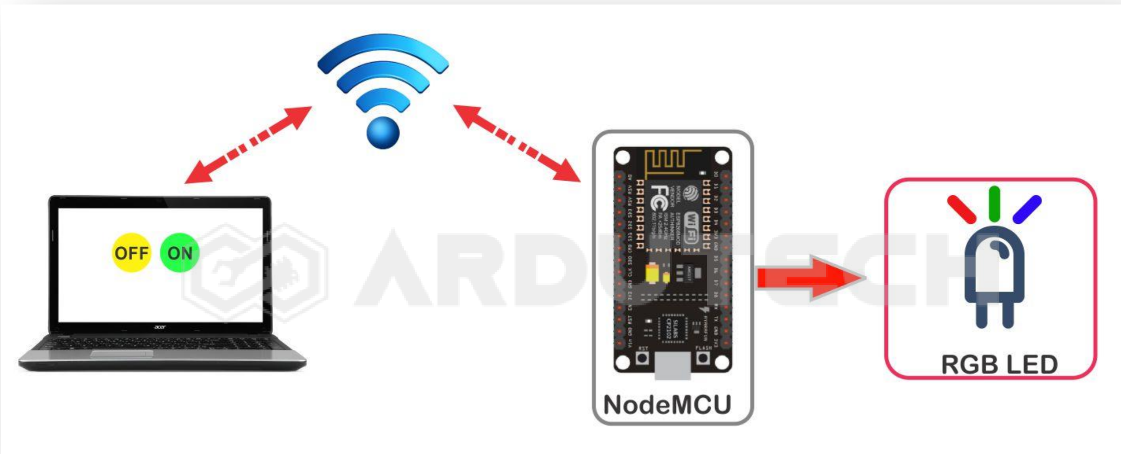
# RGB LED Strip WS2812 Controller Wabpage

## Deskripsi

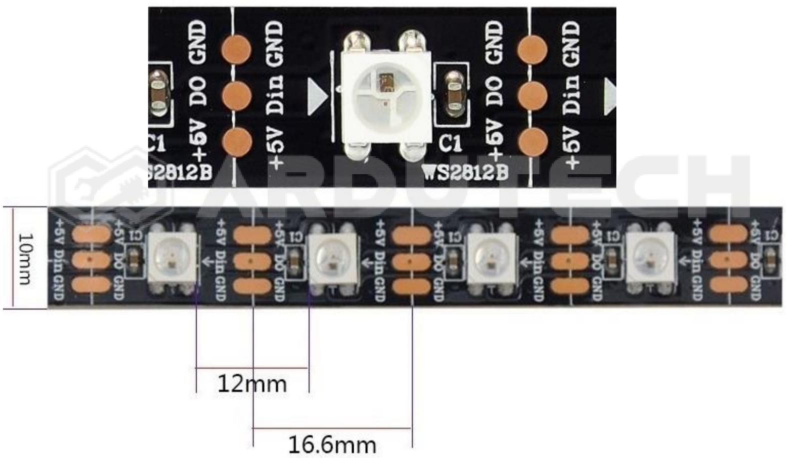
Mengontrol warna LED RGB serta kecerahan cahayanya melalui halaman web (web page). Cukup dengan memilih warna di tampilan web maka warna LED akan menyesuaikan.

## Cara Kerja

NodeMCU terhubung dengan jaringan internet melalui WiFi sehingga dapat berkomunikasi dengan melalui sebuah web page dengan IP Address yang sama. Dari web page kita dapat melakukan proses kontrol LED RGB dengan mengirim data dari web ke NodeMCU sehingga warna LED RGB dapat kita atur melalui jaringan WiFi.




Addressable RGB LED merupakan LED RGB (Red, Green, Blue) yang berbentuk strip dengan tambahan sebuah driver. Berbeda dengan LED RGB yang mempunyai pin/kaki untuk “R” kaki “G” dan kaki “B”, LED jenis ini hanya mempunyai sebuah pin yaitu “Din” saja.  Jenis yang umum dipakai RGB LED WS2812 atau Adafruit NeoPixel.



RGB LED strip dihubungkan dengan NodeMCU V3 kemudian warna LED dikontrol melalui halaman web (web page). NodeMCU dan device yang mengontrol (laptop/HP) harus satu jaringan WiFi (hotspot)

## Kebutuhan Bahan

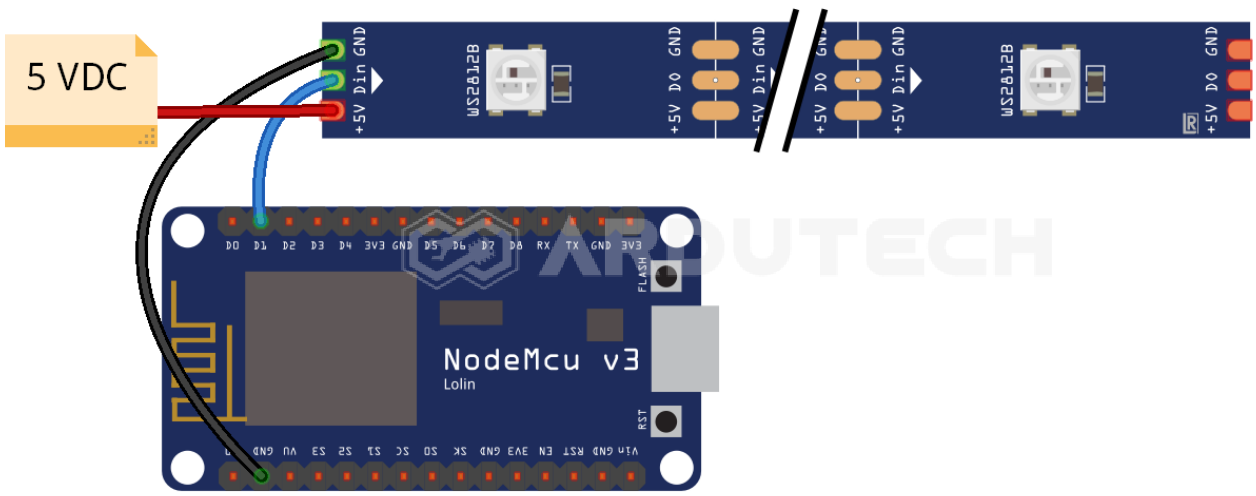
- NodeMCU V3
- Addressable RGB LED WS2812
- Kabel konektor
- Kabel micro USB

[ALAT]

## Kebutuhan Software

- Arduino IDE
- Source code html.

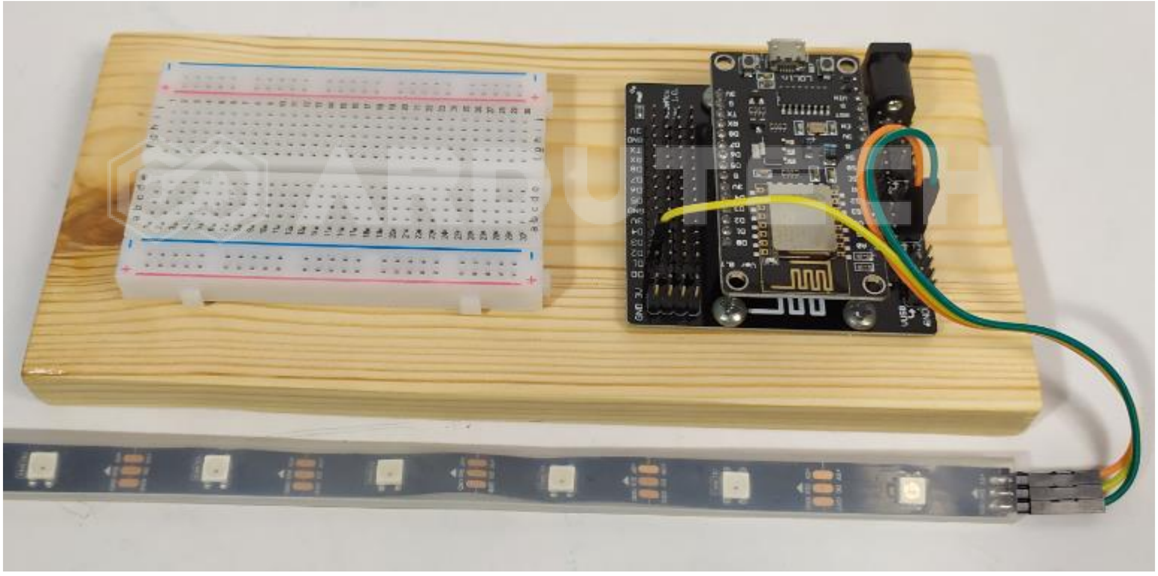
## Rangkaian/Skematik




Koneksi NodeMCU dengan RGB LED STRIP.

| NodeMCU | RGB LED Strip WS2812 |
|---|---|
| **D1** | Din |

| | |
|---|---|
| **GND** | GND |



## Program/Source Code

Program pada proyek ini memerlukan library :

- **ESP8266WiFi.h**
- **Adafruit_NeoPixel.h**

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/*************************************
 * Project RGB LED Strip WS2812 Webpage
 * Board   : NodeMCU ESP8266 V3
 * Input   : Webpage (IP Address)
 * Output  : RGB LED Strip
 * 99 Proyek IoT
 * www.ardutech.com
 *************************************/

#include <ESP8266WiFi.h>
#include <WiFiClient.h>
#include <ESP8266WebServer.h>
#include <ESP8266mDNS.h>
#include <Adafruit_NeoPixel.h>
```

```
#include <EEPROM.h>
#include "html_pages.h"

#define PIN D1
#define NUMPIXELS 3

#define CANDLEPIXELS 6
Adafruit_NeoPixel strip = Adafruit_NeoPixel(NUMPIXELS, PIN, NEO_GRB +
NEO_KHZ800);
Adafruit_NeoPixel *wick;
byte state;
unsigned long flicker_msecs;
unsigned long flicker_start;
byte index_start;
byte index_end;
//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
const char *ssid = "ArdutechWiFi";// Nama Hotspot/WiFi
const char *password = "12345678";//  Password
MDNSResponder mdns;
ESP8266WebServer server ( 80 );

const int delayval = 500;
String rgb_now = "#0000ff";
long red_int = 0;
long green_int = 0;
long blue_int = 255;
int brightness = 255;
int mode_flag = 1;

#define WICK_PIN                PIN
```

```
#define UNCONNECTED_PIN         2
#define BRIGHT                  0
#define UP                      1
#define DOWN                    2
#define DIM                     3
#define BRIGHT_HOLD             4
#define DIM_HOLD                5

#define INDEX_BOTTOM_PERCENT    10
#define INDEX_BOTTOM            128
#define INDEX_MIN               192
#define INDEX_MAX               255

#define DOWN_MIN_MSECS          20
#define DOWN_MAX_MSECS          250
#define UP_MIN_MSECS            20
#define UP_MAX_MSECS            250
#define BRIGHT_HOLD_PERCENT     20
#define BRIGHT_HOLD_MIN_MSECS   0
#define BRIGHT_HOLD_MAX_MSECS   100
#define DIM_HOLD_PERCENT        5
#define DIM_HOLD_MIN_MSECS      0
#define DIM_HOLD_MAX_MSECS      50
#define MINVAL(A,B)             (((A) < (B)) ? (A) : (B))
#define MAXVAL(A,B)             (((A) > (B)) ? (A) : (B))

void handleIndex();
void handleNotFound();
void handleSwitchOn();
void handleSwitchOff();
void handleSetColour();
```

```
void handleColour();
void handleSetBrightness();
void handleBrightness();
void handleSelectMode();
void handle_mode1();
void handle_mode2();
void handle_mode3();
void handle_mode4();
void set_color(byte);
void light_up_all();
void turn_off_all();
uint32_t Wheel(byte);
//=====================================
void setup ( void ) {
     Serial.begin ( 9600 );
     WiFi.begin ( ssid, password );
     Serial.println ( "" );
     // Wait for connection
     while ( WiFi.status() != WL_CONNECTED ) {
          delay ( 500 );
          Serial.print ( "." );
     }
  EEPROM.begin(5);
  Serial.println ( "" );
     Serial.print ( "Connected to " );
     Serial.println ( ssid );
     Serial.print ( "IP address: " );
     Serial.println ( WiFi.localIP() );
     if ( mdns.begin ( "esp8266", WiFi.localIP() ) ) {
          Serial.println ( "MDNS responder started" );
     }
```

```
  Serial.println ( "HTTP server started" );
  server.on ( "/", handleIndex );
  server.onNotFound ( handleNotFound );
  server.on ( "/switch_on", handleSwitchOn);
  server.on ( "/switch_off", handleSwitchOff);
  server.on ( "/set_colour", handleSetColour);
  server.on ( "/set_colour_hash", handleColour );
  server.on ( "/set_brightness", handleSetBrightness);
  server.on ( "/set_bright_val", handleBrightness);
  server.on ( "/select_mode", handleSelectMode);
  server.on ( "/set_mode1", handle_mode1);
  server.on ( "/set_mode2", handle_mode2);
  server.on ( "/set_mode3", handle_mode3);
  server.on ( "/set_mode4", handle_mode4);

//EEPROM Memory//
//Mem Location ||--0--||--1--||--2--||--3--||--4--||--5--||--6--||
//                red   green  blue   bright  mode

  red_int = EEPROM.read(0);
  green_int = EEPROM.read(1);
  blue_int = EEPROM.read(2);
  brightness = EEPROM.read(3);
  mode_flag = EEPROM.read(4);

     server.begin();
  strip.begin();
  randomSeed(analogRead(UNCONNECTED_PIN));
```

```
  wick = new Adafruit_NeoPixel(CANDLEPIXELS, WICK_PIN, NEO_GRB + NEO_KHZ800);
  wick->begin();
  wick->show();

  set_color(255);
  index_start = 255;
  index_end = 255;
  state = BRIGHT;
  handleSwitchOn();
}

void loop ( void ) {
     mdns.update();
     server.handleClient();
  if (mode_flag==4){

  unsigned long current_time;
  current_time = millis();
  switch (state)
    {
    case BRIGHT:
    // Serial.println("Bright");
      mdns.update();
      server.handleClient();
      flicker_msecs = random(DOWN_MAX_MSECS - DOWN_MIN_MSECS) + DOWN_MIN_MSECS;
      flicker_start = current_time;
      index_start = index_end;
      if ((index_start > INDEX_BOTTOM) &&
          (random(100) < INDEX_BOTTOM_PERCENT))
```

```
        index_end = random(index_start - INDEX_BOTTOM) + INDEX_BOTTOM;
      else
        index_end = random(index_start - INDEX_MIN) + INDEX_MIN;

      state = DOWN;
      break;

    case DIM:
      mdns.update();
      server.handleClient();
      flicker_msecs = random(UP_MAX_MSECS - UP_MIN_MSECS) + UP_MIN_MSECS;
      flicker_start = current_time;
      index_start = index_end;
      index_end = random(INDEX_MAX - index_start) + INDEX_MIN;
      state = UP;
      break;

    case BRIGHT_HOLD:
    case DIM_HOLD:
      mdns.update();
      server.handleClient();
      if (current_time >= (flicker_start + flicker_msecs))
        state = (state == BRIGHT_HOLD) ? BRIGHT : DIM;
      break;

   case UP:
   case DOWN:
     mdns.update();
     server.handleClient();
      if (current_time < (flicker_start + flicker_msecs))
```

```
        set_color(index_start    +    ((index_end    -    index_start)    *
(((current_time - flicker_start) * 1.0) / flicker_msecs)));
      else
        {
        set_color(index_end);

        if (state == DOWN)
          {
          if (random(100) < DIM_HOLD_PERCENT)
            {
            flicker_start = current_time;
            flicker_msecs       =       random(DIM_HOLD_MAX_MSECS       -
DIM_HOLD_MIN_MSECS) + DIM_HOLD_MIN_MSECS;
            state = DIM_HOLD;
            }
          else
            state = DIM;
          }
        else
          {
          if (random(100) < BRIGHT_HOLD_PERCENT)
            {
            flicker_start = current_time;
            flicker_msecs      =      random(BRIGHT_HOLD_MAX_MSECS      -
BRIGHT_HOLD_MIN_MSECS) + BRIGHT_HOLD_MIN_MSECS;
            state = BRIGHT_HOLD;
            }
          else
            state = BRIGHT;
          }
        }
```

```
      break;
    } }


  return;
  }

void handleIndex(){
Serial.println ( "Request for index page received" );
server.send ( 200, "text/html", page_contents);
}

void handleNotFound() {
  String message = "File Not Found\n\n";
  message += "URI: ";
  message += server.uri();
  message += "\nMethod: ";
  message += ( server.method() == HTTP_GET ) ? "GET" : "POST";
  message += "\nArguments: ";
  message += server.args();
  message += "\n";

  for ( uint8_t i = 0; i < server.args(); i++ ) {
    message += " " + server.argName ( i ) + ": " + server.arg ( i ) +
"\n";
  }

  server.send ( 404, "text/plain", message );
}

 void handleSwitchOn() {
```

```
  mode_flag = EEPROM.read(4);
  delay(100);
  switch(mode_flag){
      case 1:handle_mode1();
      break;
      case 2:handle_mode2();
      break;
      case 3:handle_mode3();
      break;
      case 4:handle_mode4();
      break;
      default:
           light_up_all();
      break;
  }
  server.send ( 200, "text/html", "<SCRIPT language='JavaScript'>window.location='/';</SCRIPT>" );
  };

 void handleSwitchOff() {
  mode_flag=1;
  delay(100);
  turn_off_all();
  server.send ( 200, "text/html", "<SCRIPT language='JavaScript'>window.location='/';</SCRIPT>" );
  }

void handleSetColour() {
        server.send ( 200, "text/html", colour_picker);
  }
```

```
void handleSetBrightness(){
   server.send ( 200, "text/html", bright_set);
  }

void handleSelectMode(){
  server.send ( 200, "text/html", mode_page );
  }


void handleColour(){
  char buf_red[3];
  char buf_green[3];
  char buf_blue[3];
  String message = server.arg(0);
  rgb_now = message;
  rgb_now.replace("%23", "#");
  String red_val = rgb_now.substring(1,3);
  String green_val = rgb_now.substring(3,5);
  String blue_val = rgb_now.substring(5,7);

  mode_flag=1;

  red_val.toCharArray(buf_red,3);
  green_val.toCharArray(buf_green,3);
  blue_val.toCharArray(buf_blue,3);
  red_int = gamma_adjust[strtol( buf_red, NULL, 16)];
  green_int = gamma_adjust[strtol( buf_green, NULL, 16)];
  blue_int = gamma_adjust[strtol( buf_blue, NULL, 16)];

 EEPROM.write(0,red_int);
 EEPROM.write(1,green_int);
```

```
 EEPROM.write(2,blue_int);
 EEPROM.commit();

   light_up_all();
   String java_redirect = "<SCRIPT language='JavaScript'>window.location='/set_colour?";
          java_redirect += message;
          java_redirect += "';</SCRIPT>";
   server.send ( 200, "text/html", java_redirect );
  }

 void handleBrightness(){
  String message = server.arg(0);
  String bright_val = message.substring(0,3);
  brightness =  bright_val.toInt();
  EEPROM.write(3,brightness);
  EEPROM.commit();

   String java_redirect = "<SCRIPT language='JavaScript'>window.location='/set_brightness?";
          java_redirect += brightness;
          java_redirect += "';</SCRIPT>";
   server.send ( 200, "text/html", java_redirect);
  }

  void light_up_all(){
  for(int i=0;i<NUMPIXELS;i++){
   strip.setPixelColor(i, strip.Color(brightness*red_int/255,brightness*green_int/255,brightness*blue_int/255));            // Set colour with gamma correction and brightness adjust value.
   strip.show();}
```

```
  }


  void turn_off_all(){
   mode_flag=999;
  for(int i=0;i<NUMPIXELS;i++){
// pixels.Color takes RGB values, from 0,0,0 up to 255,255,255
   strip.setPixelColor(i, strip.Color(0,0,0));
// Turn off led strip
   strip.show();}
// This sends the updated pixel color to the hardware.
  }


void handle_mode1(){
  mode_flag = 1;
  EEPROM.write(4,mode_flag);
  EEPROM.commit();
  server.send ( 200, "text/html","<SCRIPT
language='JavaScript'>window.location='/';</SCRIPT>");

  while(mode_flag==1){
          light_up_all();
          loop();
  }return;
}



void handle_mode2(){
  mode_flag = 2;
  EEPROM.write(4,mode_flag);
  EEPROM.commit();
  server.send ( 200, "text/html","<SCRIPT
language='JavaScript'>window.location='/';</SCRIPT>");
```

```
  uint16_t i, j, k;
  int wait = 10;

 while(mode_flag==2){
  for(j=0; j<256; j++) {
   loop();
    for(i=0; i<NUMPIXELS; i++) {
      loop();
      strip.setPixelColor(i, Wheel((j) & 255));
      strip.show();
    } loop();

     for(k=0; k < 200; k++){
      delay(wait);
      loop();
     }loop();}
     return;}
}

void handle_mode3(){
  mode_flag = 3;
  EEPROM.write(4,mode_flag);
  EEPROM.commit();
  server.send ( 200, "text/html","<SCRIPT
language='JavaScript'>window.location='/';</SCRIPT>");
  uint16_t i, j, k;
  int wait = 10;

 while(mode_flag==3){
  for(j=0; j < 256*5; j++) {
   loop();
```

```
    for(i=0; i < NUMPIXELS; i++) {
      loop();
      strip.setPixelColor(i,Wheel(((i * 256 / NUMPIXELS) + j) & 255));
      if(mode_flag!=3){return;}
    }
    strip.show();

    for(k=0; k < 50; k++){
     delay(wait);
     loop();
    }
 }}return;
 }


uint32_t Wheel(byte WheelPos) {
  WheelPos = 255 - WheelPos;
  if(WheelPos < 85) {
    loop();
    return   strip.Color(brightness*(255   -   WheelPos   *   3)/255,   0,
brightness*(WheelPos * 3)/255);
  }
  if(WheelPos < 170) {
    WheelPos -= 85;
    loop();
    return strip.Color(0,brightness*(WheelPos * 3)/255, brightness*(255 -
WheelPos * 3)/255);
  }
  WheelPos -= 170;
  loop();
  return  strip.Color(brightness*(WheelPos  *  3)/255,  brightness*(255  -
WheelPos * 3)/255, 0);
```

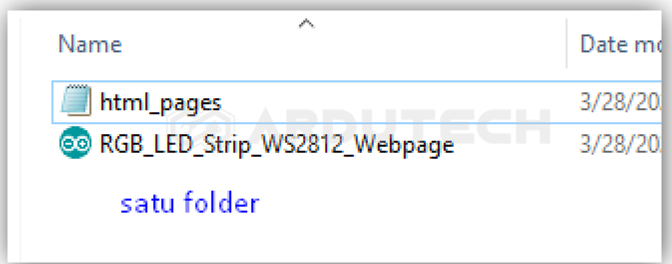
```
}



void handle_mode4(){
  mode_flag = 4;
  EEPROM.write(4,mode_flag);
  EEPROM.commit();
  server.send              (              200,              "text/html","<SCRIPT
language='JavaScript'>window.location='/';</SCRIPT>");
  turn_off_all();
  mode_flag = 4;
  loop();
  }

void set_color(byte index)
  {
  int i;
  index = MAXVAL(MINVAL(index, INDEX_MAX), INDEX_BOTTOM);
  if (index >= INDEX_MIN){
    for(i=0;i<CANDLEPIXELS;i++){
      wick->setPixelColor(i, index, (index * 3) / 8, 0);
    }}
  else if (index < INDEX_MIN){
    for(i=0;i<CANDLEPIXELS;i++){
      wick->setPixelColor(i, index, (index * 3.25) / 8, 0);
    }}

  wick->show();
  return;
  }
```

## PERHATIKAN !

**Ganti/sesuaikan variabel berikut :**

- Nama jaringan WiFi/hotspot : **ssid**
- Password jaringan WiFi/hotspot : **password**

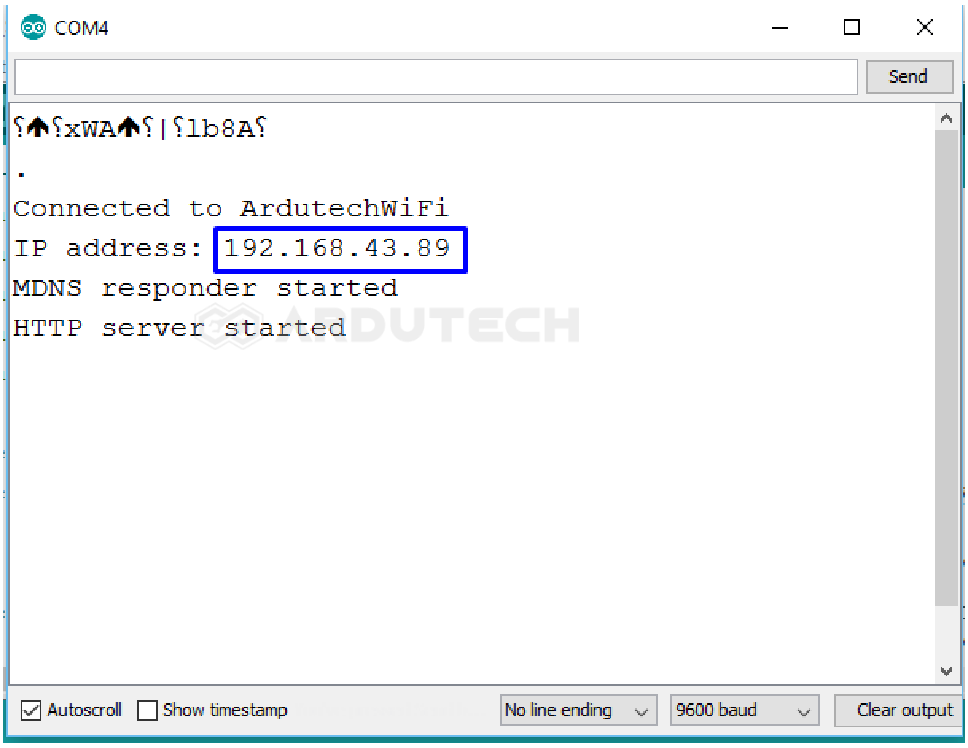
Simpan (**Save**) programnya dan **jangan** Upload dulu. Tambahkan (*copy paste*) file **html_pages.h** (sudah ada di CD) dalam satu folder dengan folder anda menyimpan file program Arduino tadi.



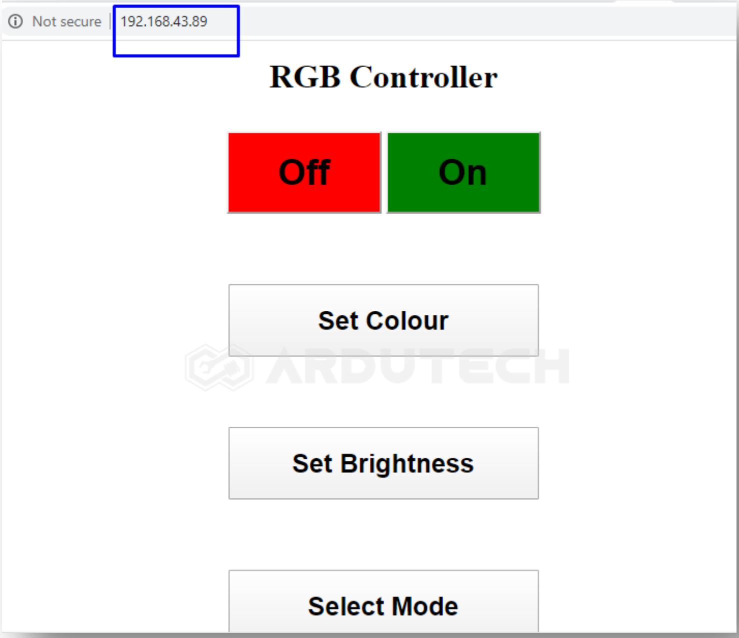

Setelah dibuat satu folder berikutnya silakan **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

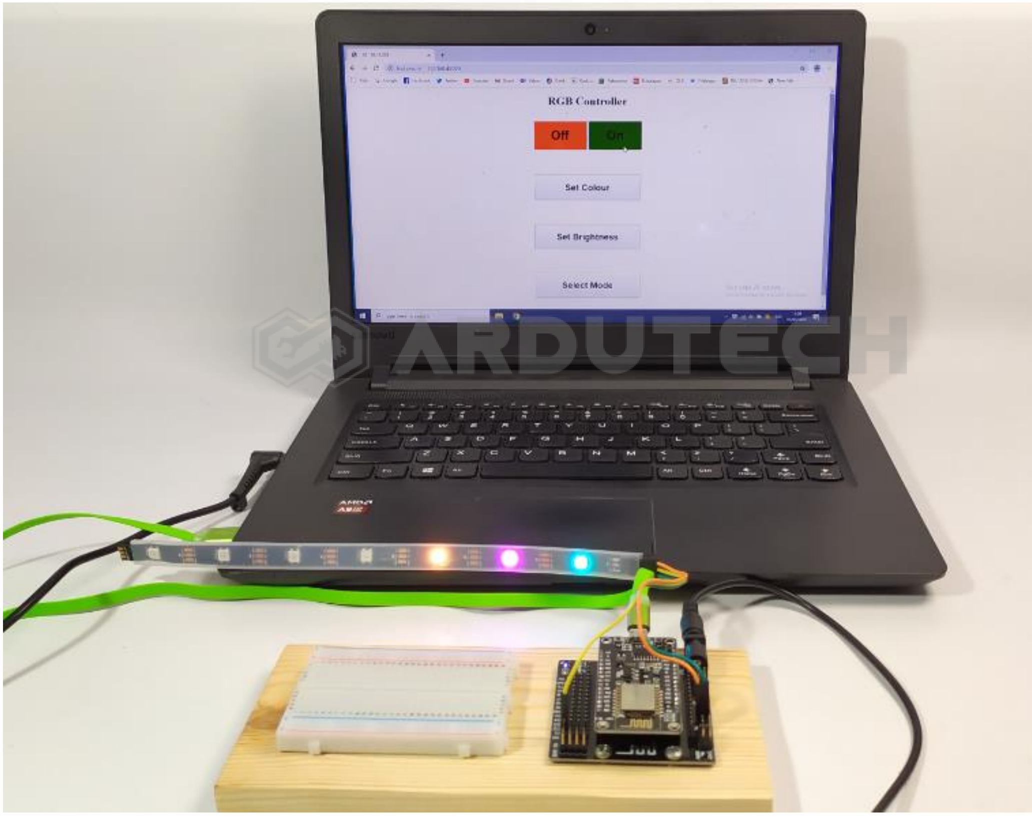
## Jalannya Alat

Setelah program berhasil di Upload, silakan buka Serial Monitor dari menu **Tools → Serial Monitor**, seting baudrate pada 9600 :




Terlihat IP address pada Serial Monitor, sekarang buka di browser anda dan tuliskan IP address tadi :



Cobalah untuk mengatur warna LED RGB dengan menekan tombol "Set Colour" kemudian pilih warnanya.

**Selamat berkarya , semoga lancar dan bermanfaat.**

**Ardutech –** *"Sahabat Inovasi Anda"*